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Course: Data structures

60% - Assessment Tool-1

1. Circular Queue (Size = 6)

Initial state:

Queue size = 5

front = -1

Rear = -1

<u>Step</u>	<u>Operation</u>	<u>Queue (Index 0-4)</u>
1.	Enqueue (20)	20 - - - -
2.	Enqueue (10)	20 10 - - -
3.	Enqueue (30)	20 10 30 - -
4.	Dequeue ()	- 10 30 - - (20 removed)
5.	Enqueue (80)	- 10 30 80 -
6.	Dequeue ()	- - 20 80 - (10 removed)
7.	Enqueue (90)	- - 30 80 90
8.	Enqueue (100)	100 - 30 80 90
9.	Dequeue ()	100 - - 80 90 (30 removed)
10.	Dequeue ()	100 - - - 90 (80 removed)
11.	Enqueue (60)	100 60 - - 90
12.	Enqueue (90)	100 60 90 - 90

Final Queue

Index	0	1	2	3	4
Value	100	60	90	-	90

Queue elements from front to Rear: 90 → 100 → 60 → 90

Final front Position : 4 (element = 90)

Final Rear Position : 2 (element = 90)

Explanation:

- Circular Queue uses modulo arithmetic to reuse empty spaces
- Enqueue inserts at the rear
- Dequeue removes from the front
- When the rear reaches the last index, it wraps around to index 0 if space is available.

2. Binary Search

Position : 1 2 3 4 5 6 7 8

Element : 66 111 212 323 432 543 643 649

Case 1: key = 111

Iteration 1

Low = 0, High = 9

Mid = 4

Element = 432

111 < 432 → search left

Iteration 2

Low = 0, High = 3

Mid = 1

element = 111 → found

Result

- element found
- Position = 2
- Comparisons = 2

Case 2: Key = 888

Iteration 1

- Mid = 4 $\rightarrow$ 432
- Search Right

Iteration 2

- Mid = 1 $\rightarrow$ 649
- Search Right

Iteration 3

- Mid = 8 $\rightarrow$ 777
- Search Right

Iteration 4

- Mid = 9 $\rightarrow$ 888
- found

Result:

- Element found
- Position = 10
- Comparisons = 4.

Case: 3 Key = 999

Iteration 1

- Mid = 4 $\rightarrow$ 432
- Search Right

Iteration 2

- Mid = 7 → 049
- Search Right

Iteration 3

- Mid = 8 → 777
- Search Right

Iteration 4

- Mid = 9 → 888
- Search Right

Now low > high so search stops

Result:

- Element Not found
- Compaction = 4